C64 USB-C PD Power Adapter v0.4 (DC only, no 50/60 Hz: the C64 TOD clock will not tick from mains)
Needs a USB-C PD charger offering 15 V. U1 (CH224A) requests 15 V via R1 = 56k.
Q1 connects the 15 V rail only after the PD contract (PG low). U2 and U3 start only above ~13.2 V (EN UVLO),
so a plain 5 V charger, or one that falls back to 9/12 V, never powers the C64.
DIN 4+5 = +5.07 V: U2 -> U4 eFuse (cuts off above 5.70 V, back on below 5.18 V; 3.3 A limit). D1 TVS.
DIN 6 = +12.3 V DC: U3 -> U5 eFuse (cuts off above 14.2 V, back on below 12.9 V; 1.5 A limit). D2 TVS, into the C64 "9 VAC" bridge.
DIN 1/2/3/7 = GND. Both eFuses also block reverse current and retry automatically after a fault.

PWR_FLAG PWR_FLAG PWR_FLAG PWR_FLAG PWR_FLAG

VBUS GND VSW 5V_BUCK 12V_BUCK

J1
USB-C (power in)
VBUS A4 B9 A9 B4
CC1 A5
CC2 B5
D- B7 A7
D+ A6 B6
SBU1 A8
SBU2 B8
SHIELD GND A1 B12
SH
GND GND

U1
CH224A
CC1 7
CC2 6
DM 5
DP 4
VBUS 8
VDD 1
GND 11
PG 10
CFG1 9
CFG2 2
CFG3 3

C1 1uF/50V
R1 56k
CFG1

R2 100k
GATE
C2 100nF/50V
Q1 AO3401A
R3 100k
PG

C3 10uF/25V
C4 100nF/50V

U2
TPS54302DDCR
VIN 3
EN 5
BOOT 6
SW 2
FB 4
GND 1

C5 100nF/50V
L1 10uH
R6 75k
R7 10k
C8 100pF

C6 22uF/25V
C7 22uF/25V

U4
TPS259470ARPWR
IN 5
EF5_EN 1
EF5_OVLO 2
DVDT 7
OUT 6
AUXOFF 3
EN/UVLO 4
FLT 9
OVLO 10
ILM
GND 8
TIMER
+5V
EF5_ILM

R13 100k
R15 75k
R14 47k
R16 20k
R17 1k

R4 470k
R5 47k

D1 SMF5.0A
D2 SMF13A
D3 GREEN
R8 1k
LED_A

C64 POWER (DIN-7 male)
+12V +5V
GND
GND +5V

U3
TPS54302DDCR
VIN 3
EN2 5
BOOT2 6
SW2 2
FB2 4
GND 1

C9 10uF/25V
C10 100nF/50V
C11 100nF/50V
L2 10uH
R11 100k
R12 5.1k
C14 100pF

C12 22uF/25V
C13 22uF/25V

U5
TPS259470ARPWR
12V_BUCK 5
EF12_EN 1
EF12_OVLO 2
DVDT 7
IN
OUT 6
AUXOFF 3
EN/UVLO
FLT 4
OVLO
ILM 9
GND 8
TIMER 10
+12V
EF12_ILM

R18 47k
R20 51k
R19 10k
R21 4.7k
R22 2.2k

USB-C PD 15 V -> C64 7-pin DIN: +5 V and +12 V DC (on the 9 VAC input), each through an OVP

Sheet: /
File: C64 USB-C PD Power Adapter.kicad_sch

Title: C64 USB-C PD Power Adapter

Size: A3	Date: 2026-11-03	**Rev: 0.4**
KiCad E.D.A. 10.0.4		Id: 1/1